Killed Noncolliding Walkers on a Finite Interval: Determinantal Spectrum, Absorption, and the Q-Process

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Abstract

For every $1 \le k \le L$, we diagonalize k continuous-time symmetric nearest-neighbour walks on $\{1, \dots, L\}$ killed at the first boundary hit or collision. The one-particle sine kernel and the Karlin–McGregor determinant yield a complete Slater basis of all $\binom{L}{k}$ chamber states. The same expansion gives the entire absorption law, every absorption-time moment, sharp leading survival asymptotics, the unique quasi-stationary law, and the conservative Doob transform with invariant density h^2.

1 Killed chamber dynamics

Fix integers $L \ge 1$ and $1 \le k \le L$. A single particle jumps one site left and right at rate one, and is killed on an attempted jump to 0 or $L+1$. The k particles have independent clocks until the first such boundary attempt or the first attempted coincidence. Before killing their ordered configuration lies in

$$\mathcal{W}_{L,k} = \{x = (x_1, \dots, x_k) : 1 \le x_1 < \cdots < x_k \le L\}.$$

For f on this chamber, extended by zero across collision and boundary faces, the sub-Markov generator is

$$(Q_k f)(x) = \sum_{a=1}^{k} \{f(x - e_a) + f(x + e_a) - 2f(x)\}. \tag{1}$$

Thus every legal one-coordinate step has rate one and $Q_k(x,x) = -2k$. This diagonal is essential: a forbidden attempt kills; it is not suppressed or reflected as in an exclusion chain.

For one particle put

$$\phi_r(j) = \sqrt{\frac{2}{L+1}} \sin \frac{\pi r j}{L+1}, \qquad \varepsilon_r = 2 - 2\cos\frac{\pi r}{L+1}, \quad 1 \le r, j \le L. \tag{2}$$

The Dirichlet difference equation and sine orthogonality give

$$p_t^{(1)}(i,j) = \frac{2}{L+1} \sum_{r=1}^{L} e^{-\varepsilon_r t} \sin\frac{\pi r i}{L+1} \sin\frac{\pi r j}{L+1}. \tag{3}$$

2 Determinant and complete spectrum

For increasing mode indices $m = (m_1 < \cdots < m_k)$ define

$$\Phi_m(x) = \det[\phi_{m_a}(x_b)]_{a,b=1}^{k}, \qquad \Lambda_m = \sum_{a=1}^{k} \varepsilon_{m_a}. \tag{4}$$

Theorem 1 (Full killed semigroup atlas)**.** *The $\binom{L}{k}$ functions Φ_m form a complete orthonormal basis of $\ell^2(\mathcal{W}_{L,k})$ and $Q_k \Phi_m = -\Lambda_m \Phi_m$. For all $t \ge 0$,*

$$P_t(x,y) = \Pr_x\{X(t) = y,\ \tau > t\} = \det[p_t^{(1)}(x_i, y_j)]_{i,j=1}^{k}, \tag{5}$$

$$P_t(x,y) = \sum_m e^{-\Lambda_m t} \Phi_m(x) \Phi_m(y). \tag{6}$$

Every eigenvalue is displayed with its multiplicity, including coincidences among distinct mode sums.

Proof. Tensor products of the one-particle eigenvectors diagonalize the generator of the independent product chain. Antisymmetrizing a tensor product gives the determinant in (4); crossing a wall changes sign, so its restriction to the ordered chamber obeys the zero boundary convention in (1). Cauchy–Binet, or direct antisymmetric tensor orthogonality, gives $\sum_{x\in\mathcal{W}_{L,k}} \Phi_m(x)\Phi_n(x) = \mathbf{1}_{m=n}$. There are exactly $\binom{L}{k} = |\mathcal{W}_{L,k}|$ modes, proving completeness and (6).

For (5), expand the determinant of (3) and apply Cauchy–Binet. Equivalently, the Karlin–McGregor path-switching involution pairs product paths at their first collision with opposite permutation sign; only never-coincident paths remain. Boundary killing is already contained in each one-particle kernel. At $t = 0$, orthonormal completeness makes either formula the identity kernel. □

The lowest mode is $m_0 = (1, \dots, k)$. Its determinant has a fixed sign. Indeed, with $\theta_b = \pi x_b/(L+1)$ and $\sin(a\theta) = \sin\theta\, U_{a-1}(\cos\theta)$,

$$h(x) := (-1)^{k(k-1)/2}\Phi_{m_0}(x) = \left(\frac{2}{L+1}\right)^{k/2} 2^{k(k-1)/2} \prod_b \sin\theta_b \prod_{i<j}(\cos\theta_i - \cos\theta_j) > 0. \tag{7}$$

Thus $\|h\|_2 = 1$ and $\Lambda_0 = \sum_{r=1}^k \varepsilon_r$ is simple.

3 Absorption, quasi-stationarity, and conditioning

Let $A_m = \sum_{y\in\mathcal{W}_{L,k}} \Phi_m(y)$, choosing the sign of m_0 so $A_0 = \sum_y h(y) > 0$.

Theorem 2 (Entire absorption law and its leading edge)**.** *For every starting state x and $t \geq 0$,*

$$S_x(t) := \Pr_x(\tau > t) = \sum_m e^{-\Lambda_m t}\Phi_m(x)A_m, \tag{8}$$

$$\Pr_x(\tau \leq t) = 1 - S_x(t), \qquad f_x(t) = \sum_m \Lambda_m e^{-\Lambda_m t}\Phi_m(x)A_m. \tag{9}$$

For every integer $r \geq 1$,

$$\mathbb{E}_x\tau^r = r!\sum_m \frac{\Phi_m(x)A_m}{\Lambda_m^r}. \tag{10}$$

If $k < L$, let

$$\Lambda_1 = \sum_{r=1}^{k-1}\varepsilon_r + \varepsilon_{k+1}, \qquad \gamma_{L,k} = \Lambda_1 - \Lambda_0 = 2\left(\cos\frac{k\pi}{L+1} - \cos\frac{(k+1)\pi}{L+1}\right).$$

Then, as $t \to \infty$,

$$S_x(t) = h(x)A_0e^{-\Lambda_0 t} + O_x(e^{-\Lambda_1 t}), \tag{11}$$

$$f_x(t) = \Lambda_0 h(x)A_0e^{-\Lambda_0 t} + O_x(e^{-\Lambda_1 t}). \tag{12}$$

The conditioned law converges from every x to the unique QSD

$$\nu(y) = \frac{h(y)}{A_0}. \tag{13}$$

Proof. Summing (6) over the terminal state proves (8). Differentiation gives (9), and integrating $r\int_0^\infty t^{r-1}S_x(t)\,\mathrm{d}t$ proves (10); every Λ_m is positive, so these finite operations are valid. Although individual spectral coefficients can have either sign, the semigroup proves the resulting survival and density are nonnegative.

The energies (2) strictly increase. Therefore the unique smallest mode is $(1, \dots, k)$ and, when $k < L$, the next energy is obtained only by replacing k with $k+1$. Separating these terms gives (11)–(12). Division of $P_t(x,y)$ by $S_x(t)$ gives (13). Finally the chamber adjacency graph is connected (successively move gaps to a packed configuration), so Perron–Frobenius makes the positive left ground eigenvector unique. Since Q_k is symmetric, that left vector is h; hence no other QSD exists. □

Proposition 1 (Immortal Doob dynamics)**.** *The ground-state transform has off-diagonal and diagonal rates*

$$q^h(x,y) = q(x,y)\frac{h(y)}{h(x)} \quad (x \neq y), \qquad q^h(x,x) = q(x,x) + \Lambda_0. \tag{14}$$

It is conservative, irreducible, and reversible with the already normalized invariant law $\pi^h(x) = h(x)^2$. For $k < L$ its relaxation gap is $\gamma_{L,k}$ above.

Proof. The identity $Q_k h = -\Lambda_0 h$ makes every row sum in (14) zero. Symmetry of q gives $h(x)^2 q^h(x,y) = h(x)h(y)q(x,y) = h(y)^2 q^h(y,x)$. Conjugation by multiplication with h shifts the killed spectrum by Λ_0, so the first positive relaxation energy is $\Lambda_1 - \Lambda_0$. □

Source ownership and AI-use statement

Karlin and McGregor [1] established the coincidence determinant for birth–death processes. Darroch and Seneta [2] treat quasi-stationarity in finite continuous-time absorbing chains. We claim no priority for those results; the contribution here is a closed source-local derivation and auditable finite certificate for the frozen interval model. AI tools assisted algebra checking, hostile mutation design, and manuscript preparation. Every theorem dependency, numerical cutoff, and nonclaim is exposed in the package.

References

[1] S. Karlin and J. McGregor, "Coincidence probabilities," *Pacific Journal of Mathematics* 9 (1959), 1141–1164, doi:10.2140/pjm.1959.9.1141.

[2] J. N. Darroch and E. Seneta, "On quasi-stationary distributions in absorbing continuous-time finite Markov chains," *Journal of Applied Probability* 4 (1967), 192–196, doi:10.2307/3212311.